

Internship Project Phase 3 Report

Name: Ashwin R Vasistha

USN: 1BO22CS006

College: Brindavan College of Engineering

Branch: CSE

e-mail: ashurashwin@gmail.com

ROLE: DA Intern

Project: E-Commerce Sales Performance Dashboard with Cohort Analysis

External guide:

- 1. Rakshit Shetty – Director, Rooman Technologies**
- 2. Shruthi M – Mentor, Rooman Technologies**

Project Repository (GitHub):

<https://github.com/ASHU-03O/E-Commerce-Sales-Analytics-Dashboard>

1. Introduction

This project focuses on analyzing e-commerce sales data using **Power BI** to gain meaningful insights into business performance, customer behavior, and retention trends. The objective is to transform raw transactional data into an interactive dashboard that supports data-driven decision-making.

The dashboard provides a comprehensive overview of revenue trends, product performance, customer segmentation, and retention patterns using advanced analytical techniques such as **Cohort Analysis**.

2. Objectives

- Analyze overall sales performance and revenue trends
- Identify top-performing products and regions
- Segment customers based on purchasing behavior
- Evaluate customer retention using cohort analysis
- Provide actionable insights to improve business strategy

3. Dataset Description

The dataset used in this project contains transactional data from an e-commerce platform.

Key Columns:

Column Name	Description
Order ID	Unique Order identifier
Customer ID	Unique identifier for customers
Revenue	Total sales value
Quantity	Number of items purchased
State	Customer location

Column Name	Description
Sub-Category	Product name

The dataset contains transactional records from an online retail platform covering multiple countries. It enables analysis of customer purchasing behavior and revenue trends over time.

4. Data Preprocessing

The following steps were performed to prepare the data:

- Removed missing and invalid values
- Converted date fields into proper format
- Created new calculated fields:
 - **CohortMonth** (first purchase month of customer)
 - **CohortIndex** (time difference in months)
- Aggregated data to compute retention rates
- Created measures for KPIs such as:
 - Total Revenue
 - Total Orders
 - Total Customers
 - Average Order Value
 - Retention Percentage

4.1 Key Measures (DAX)

- Total Revenue = SUM(Revenue)
- Total Orders = DISTINCTCOUNT(Invoice)
- Total Customers = DISTINCTCOUNT(Customer ID)
- Average Order Value = Total Revenue / Total Orders

5. Methodology

5.1 Cohort Analysis

Cohort analysis groups customers based on their first purchase month and tracks their behavior over time. This helps in understanding retention patterns and customer lifecycle.

The retention rate is calculated as the percentage of customers from the initial cohort who made repeat purchases in subsequent months.

6. Dashboard Overview

The dashboard consists of two main pages:

6.1 Page 1: Sales Performance Overview

This page provides a high-level summary of business performance:

- **KPI Cards:** Total Revenue, Sum of Profit , Total Orders, Total Customers, Average Order Value
- **Monthly Revenue Trend:** Shows sales growth over time
- **Profit Trend:** Shows profit growth over time
- **Top Products by Revenue:** Identifies best-performing products
- **Revenue by State:** Highlights geographical contribution

6.2 Page 2: Customer Retention Analysis

This page focuses on customer behavior and retention:

- **Retention Decay Curve:** Shows how customer retention changes over time
- **Cohort Heatmap:** Displays retention rates across different cohorts
- **KPIs:** Retention rates , Repeated Customers
- **Top Customers by Revenue:** Identifies Revenue by Customers
- **Customer Type:** Identifies Total Customers by Customer Type

7. Key Insights

- **Electronics is the highest revenue-generating category**
The Electronics category contributes the largest share of total revenue (around 38%), outperforming Clothing and Furniture. This indicates strong customer demand for electronic products and highlights it as the most profitable business segment.
- **Repeat customers dominate the customer base**
Approximately 67.8% of customers are repeat customers, while only 32.2% are new customers. This demonstrates strong customer retention and indicates that existing customers are a major contributor to overall business revenue.
- **Revenue shows strong growth toward the end of the period**
Monthly revenue trends reveal a noticeable increase from October 2018 onward, with peak sales occurring around January–March 2019. This suggests seasonal demand, successful promotions, or increased customer activity during that period.
- **Profitability fluctuates significantly across months**
Profit analysis shows losses during April–September, while January, March, and November generated high profits. This indicates that despite stable revenue, operational costs or discounts may have impacted profitability during mid-year months.

7.1 Hypothesis Validation

Phase 1 established four hypotheses which can now be evaluated against the findings:

- **H1 – Repeat customers contribute more revenue:** Partially supported. The analysis identified high-value repeat customers, and the cohort data confirms sustained repeat purchasing, though an exact revenue split was not isolated.
- **H2 – Retention decreases after first purchase:** Confirmed. The cohort heatmap shows retention drops from 100% across all cohorts, with continued decline thereafter.
- **H3 – A small percentage of products drive majority revenue:** Confirmed. The top 10 products contribute most of total revenue, consistent with a skewed sales distribution.
- **H4 – Certain regions contribute more to overall sales:** Confirmed. Madhya Pradesh is highest of total revenue, with all other countries contributing .

8. Conclusion

This project demonstrates how data analytics can drive strategic decisions in e-commerce by identifying growth opportunities, customer retention gaps, and revenue concentration risks.

The Power BI dashboard successfully transforms raw data into meaningful insights, helping identify key business opportunities. The analysis highlights the importance of customer retention, product optimization, and regional performance.

By leveraging these insights, businesses can:

- Improve customer engagement strategies
- Focus on high-value customers
- Optimize product offerings

9. Future Scope

- Implement predictive analytics for sales forecasting
- Develop customer churn prediction models
- Integrate real-time data for live dashboards
- Enhance personalization strategies using customer segmentation
- Implement recommendation systems for product suggestions

Screenshots of Dashboard



Figure 1: Sales Performance Dashboard (Page 1)



Figure 2: Customer Retention Analysis (Page 2)